

SAT Math Mastery · Week 4A

Quadratics II: Discriminant, Tangency & Vieta

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics number of solutions via the discriminant · tangent line to a parabola · sum and product of roots · nonlinear systems (line + parabola, line + circle) · parameter conditions · identities true for all x

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Decide the number of real solutions of $ax^2 + bx + c = 0$ from $\Delta = b^2 - 4ac$ (also from a graph).
- S2.** Find a constant that makes an equation have exactly one, two, or no real solutions (strict vs. non-strict inequalities, integer counts).
- S3.** Use Vieta: sum $-\frac{b}{a}$ and product $\frac{c}{a}$ of the solutions without solving; build symmetric expressions such as $r^2 + s^2$.
- S4.** Substitute a line into a parabola or circle, then use Δ to count intersection points or force tangency.
- S5.** Treat “intersect at two points” problems with the sum and product of the x -coordinates (midpoint, distance).
- S6.** Handle hidden conditions: leading coefficient could be 0, “distinct” vs. “real”, sign conditions on roots.
- S7.** Match coefficients when two expressions are equal for all x (identities).
- S8.** Read the sign of a , b , c and Δ from the picture of a parabola.

2 · REQUIRED FACTS

- $\Delta > 0$: two distinct real solutions. $\Delta = 0$: exactly one real (repeated) solution. $\Delta < 0$: no real solutions.
- A line is tangent to a parabola (or circle) exactly when the substituted quadratic has $\Delta = 0$: one common point.
- For $ax^2 + bx + c = 0$ with solutions r and s : $r + s = -\frac{b}{a}$ and $rs = \frac{c}{a}$. These hold whether or not the solutions are nice.
- “No real solutions” $\iff$ the graph of $y = ax^2 + bx + c$ never touches the x -axis.
- If a and c have opposite signs, then $\Delta > 0$ (the parabola must cross the x -axis).
- If the leading coefficient contains a parameter, check the value that makes it 0: the equation stops being quadratic.

3 · REQUIRED FORMULAS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad \Delta = b^2 - 4ac, \quad |r - s| = \frac{\sqrt{\Delta}}{|a|}$$

$$r + s = -\frac{b}{a}, \quad rs = \frac{c}{a}, \quad r^2 + s^2 = (r + s)^2 - 2rs, \quad r^3 + s^3 = (r + s)^3 - 3rs(r + s)$$

$$\text{vertex: } x = -\frac{b}{2a}, \quad (x - r)(x - s) = x^2 - (r + s)x + rs, \quad \text{one solution } \iff ax^2 + bx + c = a(x - h)^2$$

4 · MINI THEORY SHEET

Count solutions in three steps. (1) Move everything to one side: $ax^2 + bx + c = 0$. (2) Compute $\Delta = b^2 - 4ac$ (keep the constant k symbolic). (3) Turn the wording into an inequality on Δ .

Wording	Condition on Δ
exactly one (real) solution, tangent, touches	$\Delta = 0$
two distinct real solutions, crosses twice	$\Delta > 0$
no real solutions, never crosses / meets	$\Delta < 0$
at least one real solution	$\Delta \geq 0$
the <i>greatest integer</i> with no solution	largest integer satisfying the strict inequality

Line meets curve. Substitute the line $y = mx + k$ into the curve. The result is a quadratic in x whose Δ counts intersection points. Tangency $\iff \Delta = 0$.

Two intersection points. The x -coordinates are the two solutions, so their sum and product come straight from Vieta; the y -coordinates follow from the line.

Parameter in the leading coefficient. For $(k-1)x^2 + \dots = 0$, test $k = 1$ separately: the equation becomes linear and may have exactly one solution.

Identities. If two expressions are equal for every x , expand and match the coefficients of x^2 , x and the constants.

Reading a graph. Opens up $\Rightarrow a > 0$. Vertex right of the y -axis $\Rightarrow -\frac{b}{2a} > 0$, so a and b have opposite signs. y -intercept = c . Crosses the x -axis twice $\Rightarrow \Delta > 0$.

Speed shortcuts.

Situation	Shortcut
sum or product of solutions asked $r^2 + s^2$, $r^3 + s^3$, $\frac{r}{s} + \frac{s}{r}$	Vieta: no solving, no factoring; first rewrite as $ax^2 + bx + c = 0$ express with $r + s$ and rs
$x^2 + bx + c$ with exactly one solution	$c = \left(\frac{b}{2}\right)^2$ (perfect square)
$ r - s $ between two solutions	$\frac{\sqrt{\Delta}}{ a }$
line $y = mx + k$ and $y = x^2 + \dots$	set equal, one quadratic, then Δ
circle $x^2 + y^2 = R^2$ and line	line tangent $\iff$ distance from centre to line = R
number of integer k with $\Delta < 0$	$k^2 < N$ gives $-\sqrt{N} < k < \sqrt{N}$: count the integers carefully (strict)
$ax^2 + bx + c = 0$ with $ac < 0$	always two real solutions

5 · GUIDED EXAMPLES

Example 1 — Constant that gives exactly one solution

For what value of k does $3x^2 - 12x + k = 0$ have exactly one real solution?

$\Delta = (-12)^2 - 4(3)(k) = 144 - 12k$. Exactly one solution: $\Delta = 0 \Rightarrow k = 12$. (Check: $3x^2 - 12x + 12 = 3(x-2)^2$.)

Example 2 — Tangent line to a parabola

The line $y = 4x + k$ is tangent to the parabola $y = x^2 + 2x + 5$. Find k .

Set equal: $x^2 + 2x + 5 = 4x + k \Rightarrow x^2 - 2x + (5 - k) = 0$. Tangent $\Rightarrow \Delta = 0$: $4 - 4(5 - k) = 0 \Rightarrow k = 4$.
(The touching point: $x = 1$, $y = 8$.)

Example 3 — Vieta without solving

The solutions of $2x^2 - 9x + 3 = 0$ are r and s . Find $r^2 + s^2$.

$r + s = \frac{9}{2}$, $rs = \frac{3}{2}$. Then $r^2 + s^2 = (r + s)^2 - 2rs = \frac{81}{4} - 3 = \frac{69}{4}$.

6 · PRACTICE SET — 46 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

How many distinct real solutions does the equation $2x^2 - 5x + 4 = 0$ have?

(A) Infinitely many

(B) Zero ✓

(C) Exactly one

(D) Exactly two

1. $\Delta = b^2 - 4ac = (-5)^2 - 4(2)(4) = 25 - 32 = -7$.

2. $\Delta < 0$, so the equation has no real solutions.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

What is the sum of the solutions of $x^2 - 9x + 14 = 0$?

1. By Vieta, the sum is $-\frac{b}{a} = -\frac{-9}{1} = 9$.

2. (Check: $(x - 2)(x - 7) = 0$, and $2 + 7 = 9$.)

Answer: 9

3. EASY · NO CALCULATOR NEEDED

What is the product of the solutions of $3x^2 - 7x - 12 = 0$?

(A) $\frac{7}{3}$

(B) $-\frac{7}{3}$

(C) -4 ✓

(D) 4

1. By Vieta, the product is $\frac{c}{a} = \frac{-12}{3} = -4$.

2. The discriminant is $49 + 144 > 0$, so both solutions are real.

Answer: (C)

4. EASY · NO CALCULATOR NEEDED

The equation $x^2 + 6x + c = 0$ has exactly one real solution, where c is a constant. What is the value of c ?

1. Exactly one solution means $\Delta = 0$: $6^2 - 4(1)(c) = 0$.

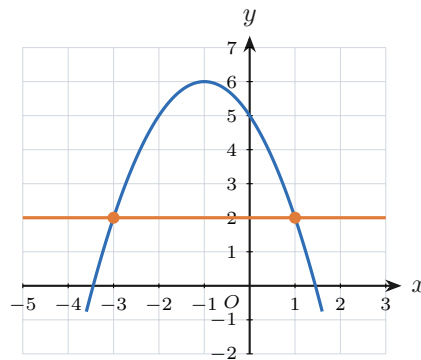
2. $36 - 4c = 0 \Rightarrow c = 9$.

Answer: 9

5.

EASY · DESMOS

The graph of a system of two equations in the xy -plane is shown. How many solutions does the system have?



(A) 2 ✓

(B) 3

(C) 0

(D) 1

1. Solutions of a system are the points where the two graphs meet.
2. The line $y = 2$ crosses the parabola at two points, $(-3, 2)$ and $(1, 2)$.
3. The system has 2 solutions.

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

For which of the following values of k does the equation $2x^2 + kx + 8 = 0$ have no real solutions?

(A) -9 (B) -8

(C) 8

(D) 5 ✓

1. $\Delta = k^2 - 4(2)(8) = k^2 - 64$.
2. No real solutions: $k^2 - 64 < 0 \Rightarrow -8 < k < 8$.
3. Only 5 lies strictly between -8 and 8 . ($k = \pm 8$ gives exactly one solution; $k = -9$ gives two.)

Answer: (D)

7.

MEDIUM · NO CALCULATOR NEEDED

The equation $2x^2 + 8x + k = 0$ has no real solutions, where k is an integer. What is the least possible value of k ?

1. $\Delta = 8^2 - 4(2)(k) = 64 - 8k$.
2. No real solutions: $64 - 8k < 0 \Rightarrow k > 8$.
3. $k = 8$ gives $\Delta = 0$ (one solution), so it is excluded. The least integer is 9.

Answer: 9

8.

MEDIUM · NO CALCULATOR NEEDED

The equation $3x^2 - 5 = 11x + x^2$ has two real solutions. What is the sum of the solutions?

(A) $-\frac{11}{2}$

(B) $-\frac{5}{2}$

(C) $\frac{11}{2}$ ✓

(D) 11

1. Move all terms to one side: $3x^2 - x^2 - 11x - 5 = 0 \Rightarrow 2x^2 - 11x - 5 = 0$.

2. Sum = $-\frac{b}{a} = -\frac{-11}{2} = \frac{11}{2}$.

Answer: (C)

9.

MEDIUM · NO CALCULATOR NEEDED

In the xy -plane, the line $y = 2x + k$ is tangent to the parabola $y = x^2 + 4x + 7$, where k is a constant. What is the value of k ?

(A) 3

(B) 6 ✓

(C) 7

(D) -6

1. Set the expressions equal: $x^2 + 4x + 7 = 2x + k \Rightarrow x^2 + 2x + (7 - k) = 0$.

2. Tangent means exactly one common point: $\Delta = 4 - 4(7 - k) = 0$.

3. $4k - 24 = 0 \Rightarrow k = 6$.

Answer: (B)

10.

MEDIUM · DESMOS

In the xy -plane, the line $y = 6$ intersects the parabola $y = x^2 - 6x + 11$ at two points. What is the distance between the two points?

(A) 5

(B) 16

(C) 2

(D) 4 ✓

1. Set $x^2 - 6x + 11 = 6 \Rightarrow x^2 - 6x + 5 = 0 \Rightarrow (x - 1)(x - 5) = 0$.

2. The points are (1, 6) and (5, 6); they lie on a horizontal line.

3. Distance = $5 - 1 = 4$.

Answer: (D)

11.

MEDIUM · DESMOS

$$y = x^2 - 5$$

$$y = 2x + 3$$

The solutions to the given system of equations are (x_1, y_1) and (x_2, y_2) . What is the value of $y_1 + y_2$?

(A) 10 ✓

(B) 12

(C) 2

(D) 8

1. Substitute: $x^2 - 5 = 2x + 3 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x - 4)(x + 2) = 0$.

2. $x = 4 \Rightarrow y = 2(4) + 3 = 11$; $x = -2 \Rightarrow y = 2(-2) + 3 = -1$.

3. $y_1 + y_2 = 11 + (-1) = 10$. (Note $x_1 + x_2 = 2$ is a different quantity.)

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

The solutions of $x^2 - 10x + c = 0$ are r and s , where $r > s$ and $r - s = 4$. What is the value of c ?

1. Vieta: $r + s = 10$. Given: $r - s = 4$.

2. Add: $2r = 14 \Rightarrow r = 7$, so $s = 3$.

3. $c = rs = 7 \cdot 3 = 21$.

Answer: 21

13.

MEDIUM · NO CALCULATOR NEEDED

A rectangular garden has a perimeter of 36 meters and an area of 80 square meters. What is the length, in meters, of the longer side of the garden?

1. Let one side be w ; the other is $18 - w$ (half the perimeter).

2. $w(18 - w) = 80 \Rightarrow w^2 - 18w + 80 = 0 \Rightarrow (w - 8)(w - 10) = 0$.

3. The sides are 8 and 10; the longer side is 10 meters.

Answer: 10

14.

HARD · NO CALCULATOR NEEDED

In the equation $5x^2 + (k - 3)x + 5 = 0$, k is a constant. For which of the following values of k does the equation have exactly one real solution?

(A) -10 (B) -7 ✓(C) 7 (D) 10

1. $\Delta = (k - 3)^2 - 4(5)(5) = (k - 3)^2 - 100$.

2. $\Delta = 0 \Rightarrow k - 3 = \pm 10 \Rightarrow k = 13$ or $k = -7$.

3. Only -7 appears among the choices. ($k = \pm 10$ comes from forgetting the -3 ; $k = 7$ gives $(4)^2 - 100 \neq 0$.)

Answer: (B)

15.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $y = mx - 5$, where $m > 0$, is tangent to the parabola $y = 2x^2 + 3$. What is the value of m ?

1. Set equal: $2x^2 + 3 = mx - 5 \Rightarrow 2x^2 - mx + 8 = 0$.

2. Tangent: $\Delta = m^2 - 4(2)(8) = 0 \Rightarrow m^2 = 64 \Rightarrow m = \pm 8$.

3. Since $m > 0$, $m = 8$.

Answer: 8

16.

HARD · NO CALCULATOR NEEDED

The solutions of $x^2 - 7x + 5 = 0$ are r and s . What is the value of $r^2 + s^2$?

(A) 29

(B) 59

(C) 39 ✓

(D) 49

1. Vieta: $r + s = 7$ and $rs = 5$.

2. $r^2 + s^2 = (r + s)^2 - 2rs = 49 - 10 = 39$.

Answer: (C)

17.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $y = x + k$, where $k > 0$, is tangent to the circle $x^2 + y^2 = 50$. What is the value of k ?

1. Substitute: $x^2 + (x + k)^2 = 50 \Rightarrow 2x^2 + 2kx + (k^2 - 50) = 0$.
2. Tangent: $\Delta = (2k)^2 - 4(2)(k^2 - 50) = 400 - 4k^2 = 0 \Rightarrow k^2 = 100$.
3. Since $k > 0$, $k = 10$. (Check: distance from the origin to $x - y + 10 = 0$ is $\frac{10}{\sqrt{2}} = 5\sqrt{2} = \sqrt{50}$.)

Answer: 10

18.

HARD · NO CALCULATOR NEEDED

For how many integer values of k does the equation $3x^2 + kx + 12 = 0$ have no real solutions?

(A) 23 ✓

(B) 24

(C) 11

(D) 22

1. $\Delta = k^2 - 4(3)(12) = k^2 - 144 < 0 \Rightarrow -12 < k < 12$.
2. The integers strictly between -12 and 12 are $-11, -10, \dots, 11$.
3. That is $11 + 1 + 11 = 23$ values. (The endpoints ± 12 give $\Delta = 0$ and are excluded.)

Answer: (A)

19.

HARD · NO CALCULATOR NEEDED

The equation $x^2 - 9x + k = 0$ has two real solutions, and one solution is twice the other. What is the value of k ?

1. Let the solutions be r and $2r$. Then $r + 2r = 9 \Rightarrow r = 3$.
2. The solutions are 3 and 6.
3. $k = 3 \cdot 6 = 18$.

Answer: 18

20.

HARD · NO CALCULATOR NEEDED

In the equation $ax^2 + bx + c = 0$, a , b , and c are constants with $a > 0$ and $c < 0$. Which of the following must be true?

- I. The equation has two distinct real solutions.
- II. The two solutions are both positive.
- III. The product of the two solutions is negative.

(A) I, II, and III

(B) I only

(C) III only

(D) I and III only ✓

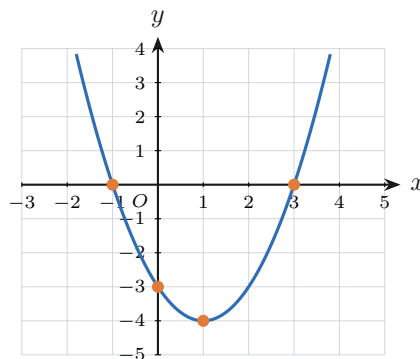
1. $ac < 0$, so $\Delta = b^2 - 4ac = b^2 + (\text{positive}) > 0$: statement I is true.
2. Product of solutions $= -\frac{c}{a} < 0$: statement III is true.
3. A negative product means the solutions have opposite signs, so II is false.
4. Only I and III must be true.

Answer: (D)

21.

HARD · DESMOS

The graph of $y = f(x)$ is shown in the xy -plane. For which of the following values of k does the equation $f(x) + k = 0$ have no real solutions?



(A) -4

(B) 4

(C) 5 ✓

(D) -5

1. The vertex $(1, -4)$ shows the minimum value of f is -4 , so $f(x)$ takes every value ≥ -4 and no value below -4 .
2. $f(x) + k = 0 \iff f(x) = -k$. No real solutions requires $-k < -4$, i.e. $k > 4$.
3. Check: $k = 5$ gives $f(x) = -5 < -4$ (none); $k = 4$ gives $f(x) = -4$ (one solution); $k = -4$ and $k = -5$ give $f(x) = 4$ and 5 (two solutions each).

Answer: (C)

22.

HARD · CALCULATOR

A ball is thrown upward from a platform. Its height, in feet, t seconds after it is thrown is $h(t) = -16t^2 + 64t + 6$. The ball is at a height of exactly H feet at exactly one time during its flight, where $H > 6$. What is the value of H ?

1. Height H at one time only: $-16t^2 + 64t + 6 = H \Rightarrow 16t^2 - 64t + (H - 6) = 0$ has $\Delta = 0$.
2. $\Delta = 64^2 - 4(16)(H - 6) = 4096 - 64(H - 6) = 0 \Rightarrow H - 6 = 64$.
3. $H = 70$. (This is the maximum height: vertex at $t = 2$, $h(2) = -64 + 128 + 6 = 70$.)

Answer: 70**23.**

HARD · NO CALCULATOR NEEDED

The equation $(ax + 3)(2x - b) = 6x^2 + cx - 12$ is true for all values of x , where a , b , and c are constants. What is the value of c ?

1. Expand the left side: $2ax^2 - abx + 6x - 3b = 2ax^2 + (6 - ab)x - 3b$.
2. Match x^2 : $2a = 6 \Rightarrow a = 3$. Match constants: $-3b = -12 \Rightarrow b = 4$.
3. Match x : $c = 6 - ab = 6 - 12 = -6$.

Answer: -6**24.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For a constant k , the equation $(k - 1)x^2 - 6x + (k - 1) = 0$ has exactly one real solution. What is the sum of all possible values of k ?

1. If $k \neq 1$ the equation is quadratic with $\Delta = 36 - 4(k - 1)^2$. Exactly one solution: $(k - 1)^2 = 9 \Rightarrow k - 1 = \pm 3 \Rightarrow k = 4$ or $k = -2$.
2. If $k = 1$ the equation becomes $-6x = 0$, which is linear and has exactly one solution, $x = 0$. So $k = 1$ also works.
3. Sum of all possible values: $4 + (-2) + 1 = 3$.

Answer: 3

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

The solutions of $x^2 - 6x + 4 = 0$ are r and s . What is the value of $r^3 + s^3$?

(A) 108

(B) 144 ✓

(C) 216

(D) 72

1. Vieta: $r + s = 6$ and $rs = 4$.

2. $r^3 + s^3 = (r + s)^3 - 3rs(r + s)$.

3. $= 6^3 - 3(4)(6) = 216 - 72 = 144$.

Answer: (B)

26.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = x + 2$ intersects the circle $x^2 + y^2 = 20$ at points P and Q . What is the value of $(PQ)^2$?

1. Substitute: $x^2 + (x + 2)^2 = 20 \Rightarrow 2x^2 + 4x - 16 = 0 \Rightarrow x^2 + 2x - 8 = 0 \Rightarrow (x + 4)(x - 2) = 0$.

2. $P = (2, 4)$ and $Q = (-4, -2)$.

3. $(PQ)^2 = (2 - (-4))^2 + (4 - (-2))^2 = 36 + 36 = 72$.

Answer: 72

27.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = 2x - 3$ is tangent to the parabola $y = x^2 - 6x + c$, where c is a constant. The point of tangency is (p, q) . What is the value of $p + q$?

(A) 5

(B) 13

(C) 4

(D) 9 ✓

1. Set equal: $x^2 - 6x + c = 2x - 3 \Rightarrow x^2 - 8x + (c + 3) = 0$.

2. Tangent: $\Delta = 64 - 4(c + 3) = 0 \Rightarrow c = 13$. The double solution is $x = \frac{8}{2} = 4$, so $p = 4$.

3. $q = 2(4) - 3 = 5$, so $p + q = 9$. (Check on the parabola: $16 - 24 + 13 = 5$.)

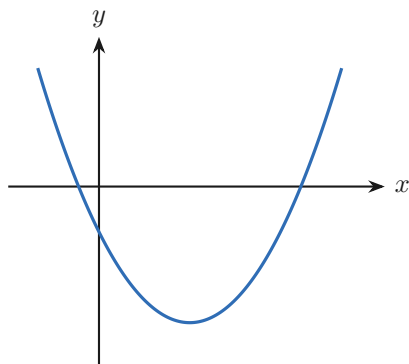
Answer: (D)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of $y = ax^2 + bx + c$ is shown in the xy -plane, where a , b , and c are constants. Which of the following must be true?

- I. $b^2 - 4ac > 0$
- II. $ab < 0$
- III. $c > 0$

**(A) I and II only ✓**

(B) II and III only

(C) I, II, and III

(D) I only

1. The parabola opens upward, so $a > 0$.
2. It crosses the x -axis twice, so $\Delta = b^2 - 4ac > 0$: I is true.
3. The vertex lies to the right of the y -axis: $-\frac{b}{2a} > 0$ with $a > 0$ gives $b < 0$, so $ab < 0$: II is true.
4. The graph crosses the y -axis below the origin, so $c < 0$: III is false. Only I and II are true.

Answer: (A)

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

What is the least integer k for which the equation $x^2 - kx + (k + 3) = 0$ has two distinct positive real solutions?

(A) 6

(B) 7 ✓

(C) 8

(D) 3

1. Two distinct real solutions: $\Delta = k^2 - 4(k + 3) = k^2 - 4k - 12 = (k - 6)(k + 2) > 0 \Rightarrow k > 6$ or $k < -2$.
2. Both solutions positive: sum $= k > 0$ and product $= k + 3 > 0$. Together with $\Delta > 0$ this forces $k > 6$.
3. The least integer is $k = 7$. ($k = 6$ gives $\Delta = 0$, a single repeated solution $x = 3$, so it fails the word “distinct”.)

Answer: (B)**30.**

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = 2x + k$ intersects the parabola $y = x^2 - 4x + 1$ at two points whose x -coordinates differ by 8. What is the value of k ?

1. Set equal: $x^2 - 4x + 1 = 2x + k \Rightarrow x^2 - 6x + (1 - k) = 0$.
2. The x -coordinates x_1, x_2 satisfy $x_1 + x_2 = 6$ and $x_1x_2 = 1 - k$. Given $|x_1 - x_2| = 8$: $(x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1x_2$.
3. $64 = 36 - 4(1 - k) \Rightarrow 4(1 - k) = -28 \Rightarrow 1 - k = -7 \Rightarrow k = 8$. (Check: $x^2 - 6x - 7 = 0$ gives $x = -1, 7$.)

Answer: 8

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

For real numbers p and q with $p^2 + 4q^2 = 72$, the product $(2x + p)(x + q)$ is equivalent to $2x^2 + bx + c$, where b and c are constants. What is the greatest possible value of c ?

1. Expand: $(2x + p)(x + q) = 2x^2 + (2q + p)x + pq$, so $c = pq$.
2. Write $p^2 + 4q^2 = p^2 + (2q)^2$. Since $(p - 2q)^2 \geq 0$, we have $p^2 + (2q)^2 \geq 2p(2q) = 4pq$.
3. Therefore $4pq \leq 72$, so $pq \leq 18$.
4. Equality needs $p = 2q$: $p = 6$, $q = 3$ gives $36 + 36 = 72$ and $pq = 18$. The greatest value of c is 18.

Answer: 18**32.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = mx - 5$ intersects the parabola $y = x^2 - 6x + 11$ at exactly one point, where m is a negative constant. What is the value of m ?

(A) 2

(B) 14

(C) -14 ✓

(D) -2

1. Set the expressions equal: $x^2 - 6x + 11 = mx - 5 \Rightarrow x^2 - (6 + m)x + 16 = 0$.
2. Exactly one intersection means discriminant 0: $(6 + m)^2 - 4(1)(16) = 0 \Rightarrow (6 + m)^2 = 64$.
3. $6 + m = 8$ or $6 + m = -8$, so $m = 2$ or $m = -14$.
4. The constant m must be negative, so $m = -14$. (Then $x = \frac{6 + m}{2} = -4$, and both sides equal 51 there.)

Answer: (C)

33.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $x + y = 4$ intersects the circle $x^2 + y^2 - 4x + 2y = 24$ at two points. What is the distance between these two points?

(A) $7\sqrt{2}$ ✓

(B) 14

(C) $\frac{7\sqrt{2}}{2}$

(D) 7

1. Substitute $y = 4 - x$: $x^2 + (4 - x)^2 - 4x + 2(4 - x) = 24$.
2. Expand: $2x^2 - 8x + 16 - 4x + 8 - 2x = 24 \Rightarrow 2x^2 - 14x = 0 \Rightarrow 2x(x - 7) = 0$.
3. $x = 0$ gives $(0, 4)$; $x = 7$ gives $(7, -3)$.
4. Distance = $\sqrt{(7 - 0)^2 + (-3 - 4)^2} = \sqrt{98} = 7\sqrt{2}$. (The horizontal difference 7 alone is not the distance, and no halving is needed.)

Answer: (A)

34.

SAT CHALLENGE · DESMOS

$$x^2 + y^2 = 65$$

$$x - y = 7$$

The solution (a, b) to the system of equations above satisfies $a > 0$. What is the value of $a + 3b$?

1. Write $y = x - 7$ and substitute: $x^2 + (x - 7)^2 = 65 \Rightarrow 2x^2 - 14x - 16 = 0 \Rightarrow x^2 - 7x - 8 = 0$.
2. Factor: $(x - 8)(x + 1) = 0$, so $x = 8$ or $x = -1$.
3. $x = 8$ gives $y = 1$; $x = -1$ gives $y = -8$. Only $(8, 1)$ has $a > 0$.
4. $a + 3b = 8 + 3(1) = 11$.

Answer: 11

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

For integers a , b , c , and d , $(ax + b)(cx + d) = 6x^2 + kx + 5$ for all values of x , where k is a constant. Which of the following must be true?

I. k is oddII. When k is divided by 4, the remainder is 1III. k is not divisible by 5

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only ✓1. Matching coefficients: $ac = 6$, $bd = 5$, $k = ad + bc$.2. Since $bd = 5$ is odd, b and d are both odd. Since $ac = 6$ is not divisible by 4, exactly one of a, c is even. Then $k = ad + bc \equiv a + c \equiv 1 \pmod{2}$. So I is true.3. Since $bd = 5$, exactly one of b, d is a multiple of 5; say $5 \mid b$. Then $k \equiv ad \pmod{5}$, and $a \mid 6$ and $d = \pm 1$ are not multiples of 5. (Same if $5 \mid d$.) So III is true.4. II: $(a, c, b, d) = (1, 6, 5, 1)$ gives $k = 1 \cdot 1 + 5 \cdot 6 = 31$, remainder 3 on division by 4. So II is false, even though $k = 13$ and $k = 17$ satisfy it.5. Possible values: $k = \pm 11, \pm 13, \pm 17, \pm 31$. Only I and III must be true.**Answer: (D)****36.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = k(x - 1) - 2$, where k is a positive constant, is tangent to the parabola $y = x^2 + 2x + 4$. What is the value of k ?

(A) 4

(B) 8

(C) 10 ✓(D) -2 1. Set equal: $x^2 + 2x + 4 = kx - k - 2 \Rightarrow x^2 + (2 - k)x + (k + 6) = 0$.2. Tangency means exactly one solution: $(2 - k)^2 - 4(k + 6) = 0$.3. $k^2 - 8k - 20 = 0 \Rightarrow (k - 10)(k + 2) = 0 \Rightarrow k = 10$ or $k = -2$.4. Since $k > 0$, $k = 10$. (The tangent point is $x = \frac{k - 2}{2} = 4$, which is not the answer.)**Answer: (C)**

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the graph of the quadratic function p and the graph of the line $y = 5 - x$ intersect at exactly two points, one with x -coordinate 1 and one with x -coordinate 4. If $p(0) = -3$, what is the value of $p(6)$?

1. At the intersections $p(x) = 5 - x$, so $p(x) - (5 - x)$ has zeros 1 and 4. Write $p(x) = 5 - x + a(x - 1)(x - 4)$.
2. Use $p(0) = -3$: $5 + a(-1)(-4) = -3 \Rightarrow 4a = -8 \Rightarrow a = -2$.
3. $p(6) = 5 - 6 - 2(5)(2) = -1 - 20 = -21$.
4. (Check by expanding: $p(x) = -2x^2 + 9x - 3$, and $p(6) = -72 + 54 - 3 = -21$.)

Answer: -21 **38.**

SAT CHALLENGE · DESMOS

In the xy -plane, the graphs of $y = x^2 + 6x + 4$ and $y = mx - 4$ have no point in common, where m is a constant. What is the sum of all integer values of m ?

1. Set the expressions equal: $x^2 + 6x + 4 = mx - 4 \Rightarrow x^2 + (6 - m)x + 8 = 0$.
2. No common point means no real roots: $\Delta = (6 - m)^2 - 32 < 0$, so $|m - 6| < \sqrt{32} \approx 5.66$.
3. Since $5^2 = 25 < 32 < 36 = 6^2$, $|m - 6| \leq 5$, i.e. $1 \leq m \leq 11$. (At $m = 0$ or $m = 12$ we get $|m - 6| = 6$, which is too large.)
4. Integers 1 through 11 sum to $\frac{11 \cdot 12}{2} = 66$.

Answer: 66

39.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$(x + p)(x + q) = x^2 + bx + c$$

In the equation above, p and q are real constants and b and c are integers. Which of the following must be true?

- I. $p + q$ is an integer
- II. $p^2 + q^2$ is an integer
- III. $\frac{p^2 + q^2}{2}$ is an integer

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. Expand: $x^2 + (p + q)x + pq$, so $p + q = -b$ and $pq = c$.
2. I: $p + q = -b$ is an integer. True.
3. II: $p^2 + q^2 = (p + q)^2 - 2pq = b^2 - 2c$, an integer. True.
4. III: $\frac{b^2 - 2c}{2}$ is an integer only when b^2 is even. Counterexample: $b = 3$, $c = 1$ gives $x^2 + 3x + 1$ with real roots $p, q = \frac{-3 \pm \sqrt{5}}{2}$, and $\frac{p^2 + q^2}{2} = \frac{9 - 2}{2} = \frac{7}{2}$. Not necessarily true.
5. Only I and II must be true.

Answer: (B)

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 = 45$$

$$y = mx + 15$$

In the system above, m is a negative constant. If the system has exactly one solution (x, y) , what is the value of x ?

(A) -6 (B) -3 (C) 3 **(D) 6 ✓**

1. Substitute: $x^2 + (mx + 15)^2 = 45 \Rightarrow (1 + m^2)x^2 + 30mx + 180 = 0$.
2. Exactly one solution means $\Delta = 0$: $900m^2 - 4(1 + m^2)(180) = 0 \Rightarrow 900m^2 - 720 - 720m^2 = 0 \Rightarrow 180m^2 = 720 \Rightarrow m^2 = 4$.
3. The condition $m < 0$ gives $m = -2$ (not $m = 2$).
4. The quadratic becomes $5x^2 - 60x + 180 = 0$, i.e. $x^2 - 12x + 36 = 0$, so $(x - 6)^2 = 0$ and $x = 6$.
5. Check: $y = -2(6) + 15 = 3$ and $6^2 + 3^2 = 45$. The value of x is 6 (the choice $m = 2$ would give $x = -6$).

Answer: (D)

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$y = 3x^2 + 4$$

$$y = 1 - kx$$

In the system above, k is a constant. For which of the following values of k does the system have two distinct solutions (x, y) whose x -coordinates are both negative?

(A) 7 ✓

(B) -7

(C) 2

(D) 6

1. Set the expressions equal: $3x^2 + 4 = 1 - kx \Rightarrow 3x^2 + kx + 3 = 0$.
2. Two distinct real solutions: $\Delta = k^2 - 36 > 0$, so $|k| > 6$. This excludes $k = 2$ and the tangent case $k = 6$.
3. Both roots negative: the sum $-\frac{k}{3} < 0$ requires $k > 0$. (The product $\frac{3}{3} = 1 > 0$ already makes the roots the same sign.)
4. Together: $k > 6$. Of the options only $k = 7$ qualifies; $k = -7$ gives two distinct solutions that are both positive.

Answer: (A)

42.

SAT CHALLENGE · CALCULATOR

In the xy -plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2 ?

1. Substitute $y = mx + 10$: $x^2 + (mx + 10)^2 = 12 \Rightarrow (1 + m^2)x^2 + 20mx + 88 = 0$.
2. One intersection point means the discriminant is 0: $(20m)^2 - 4(1 + m^2)(88) = 0$.
3. $400m^2 - 352 - 352m^2 = 0 \Rightarrow 48m^2 = 352 \Rightarrow m^2 = \frac{352}{48} = \frac{22}{3}$.
4. Check by distance: $\frac{10}{\sqrt{1 + m^2}} = \frac{10}{\sqrt{25/3}} = \sqrt{12}$, the radius.
5. The question asks for m^2 , so the answer is $\frac{22}{3}$ (about 7.33), not m .

Answer: $\frac{22}{3}$

43.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the line $y = kx + 4$, where k is a constant, is tangent to the parabola $y = x^2 + 4x + 13$ for exactly two values of k . For each of these values the line touches the parabola at one point. What is the greatest y -coordinate of these two points of tangency?

(A) 13

(B) 34 ✓

(C) 3

(D) 10

1. Set the expressions equal: $x^2 + 4x + 13 = kx + 4 \Rightarrow x^2 + (4 - k)x + 9 = 0$.
2. Tangent means exactly one solution, so the discriminant is 0: $(4 - k)^2 - 36 = 0 \Rightarrow 4 - k = \pm 6 \Rightarrow k = -2$ or $k = 10$.
3. The double root is $x = -\frac{4 - k}{2}$: for $k = 10$, $x = 3$; for $k = -2$, $x = -3$.
4. y -coordinates (use the parabola): $y(3) = 9 + 12 + 13 = 34$ and $y(-3) = 9 - 12 + 13 = 10$. Check with the line: $10(3) + 4 = 34$ and $-2(-3) + 4 = 10$.
5. The greater y -coordinate is 34.

Answer: (B)

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $g(x) = 2x^2 + bx + c$, where b and c are constants, has zeros $k + 1$ and $3k - 5$, where k is a constant. If $g(2) = g(10)$, what is the value of c ?

(A) 24

(B) 35

(C) 70 ✓

(D) 140

1. $g(2) = g(10)$ puts the axis of symmetry at $x = \frac{2 + 10}{2} = 6$.
2. The axis is also the midpoint of the zeros: $\frac{(k + 1) + (3k - 5)}{2} = 2k - 2 = 6 \Rightarrow k = 4$. The zeros are 5 and 7.
3. $g(x) = 2(x - 5)(x - 7) = 2x^2 - 24x + 70$, so $c = 70$.
4. Check: $g(2) = 8 - 48 + 70 = 30$ and $g(10) = 200 - 240 + 70 = 30$.

Answer: (C)

45.

SAT CHALLENGE · NO CALCULATOR NEEDED

For nonzero real numbers p and q , the equation $px^2 + 3qx + 4p = 0$ has exactly one real solution, and $pq = -48$. If $p > 0$, what is the value of $p + q$?

(A) -2 ✓(B) 2 (C) 14 (D) -14

1. Exactly one real solution means the discriminant is 0: $(3q)^2 - 4(p)(4p) = 0$, so $9q^2 = 16p^2$.

2. Taking square roots: $3q = 4p$ or $3q = -4p$, i.e. $q = \frac{4p}{3}$ or $q = -\frac{4p}{3}$.

3. If $q = \frac{4p}{3}$ then $pq = \frac{4p^2}{3} > 0$, contradicting $pq = -48$. So $q = -\frac{4p}{3}$.

4. Then $pq = -\frac{4p^2}{3} = -48 \Rightarrow p^2 = 36$. Since $p > 0$, $p = 6$ and $q = -8$.

5. $p + q = 6 - 8 = -2$. (Option 14 comes from the rejected case $q = 8$.)

Answer: (A)

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

Positive real numbers x and y satisfy $x \neq y$, $x + y = 15$ and $xy = 36$. What is the value of

$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x^2} - \frac{1}{y^2}} ?$$

(A) $\frac{5}{12}$ (B) 1 (C) 4 (D) $\frac{12}{5}$ ✓

1. Factor the denominator as a difference of squares: $\frac{1}{x^2} - \frac{1}{y^2} = \left(\frac{1}{x} - \frac{1}{y}\right)\left(\frac{1}{x} + \frac{1}{y}\right)$.

2. Since $x \neq y$, $\frac{1}{x} - \frac{1}{y} \neq 0$ and cancels, leaving $\frac{1}{\frac{1}{x} + \frac{1}{y}} = \frac{1}{\frac{x+y}{xy}} = \frac{xy}{x+y}$.

3. $\frac{xy}{x+y} = \frac{36}{15} = \frac{12}{5}$. (Check with $x = 12$, $y = 3$: $\frac{\frac{1}{12} - \frac{1}{3}}{\frac{1}{144} - \frac{1}{9}} = \frac{-\frac{1}{4}}{-\frac{5}{48}} = \frac{12}{5}$.)

4. (Option $\frac{5}{12}$ is the reciprocal.)

Answer: (D)